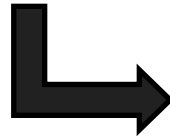




From lecture 14: We briefly learned what is primary production and what type organisms are responsible for the primary productivity in the ocean. Here we will dig into the concept of photosynthetic primary production to get some perspective of the role of the oceans to the global carbon cycle. Additionally, we will see how challenging and complex environmental studies can be.

## Lecture Session 15

1. Primary producers and primary production;
2. Study case: Karl et al. (1995);



Activity designed by: Saulo Muller  
Soares, Viktoria C Steck, Tiago  
Carrilho Bilo, Chen Chen

# What is the estimated contribution to primary production globally?

i) A 10%, B 20%, C 50%, D 80%



Amazon rainforest

ii) A 10%, B 20%, C 50%, D 80%



Pacific Ocean

TINY BUT POWERFUL



# Primary production review

What is correct about photosynthesis and primary productivity? Check all that apply.

- A** Since photosynthesis and primary productivity entail the conversion of atmospheric carbon dioxide to organic compounds, both terms are synonyms.
- B** Photosynthesis is the light-driven conversion of water and carbon dioxide to oxygen and organic carbon.
- C** Primary productivity is the rate by which organic carbon is generated.
- D** Primary productivity is the amount of biomass created by photosynthetic organisms.
- E** The majority of primary productivity stems from, but is not limited to, photosynthetic organisms.

# Primary production review

What is correct about photosynthesis and primary productivity? Check all that apply.

We can't forget about the chemosynthesis

~~A Since photosynthesis and primary productivity entail the conversion of atmospheric carbon dioxide to organic compounds, both terms are synonyms.~~

**B** Photosynthesis is the light-driven conversion of water and carbon dioxide to oxygen and organic carbon.

**C** Primary productivity is the rate by which organic carbon is generated.

~~D Primary productivity is the amount of biomass created by photosynthetic organisms.~~

**E** The majority of primary productivity stems from, but is not limited to, photosynthetic organisms.

# Primary production review

What is correct about respiration and primary productivity? Check all that apply.

**A** Respiration is the net reverse process of primary production.

**B** Respiration is the conversion of oxygen and organic carbon to water, carbon dioxide and chemical energy.

**C** Respiration is usually exhibited by heterotrophic organisms, and primary production by autotrophic organisms.

**D** All organisms respire. Net primary productivity is the primary productivity minus the carbon lost to respiration.

# Primary production review

What is correct about respiration and primary productivity? Check all that apply.

Chemosynthesis, for example, involves different reactions.

~~A Respiration is the net reverse process of primary production.~~

**B** Respiration is the conversion of oxygen and organic carbon to water, carbon dioxide and chemical energy.

~~C Respiration is usually exhibited by heterotrophic organisms, and primary production by autotrophic organisms.~~

**D** All organisms respire. Net primary productivity is the primary productivity minus the carbon lost to respiration.

# Primary production review: Study Case

Read the abstract of the paper Karl et al. 1995. Discuss with your neighbor if that changed your opinion on the contribution of trees and plankton to primary production. If so, reason why.

## **Ecosystem changes in the North Pacific subtropical gyre attributed to the 1991–92 El Niño**

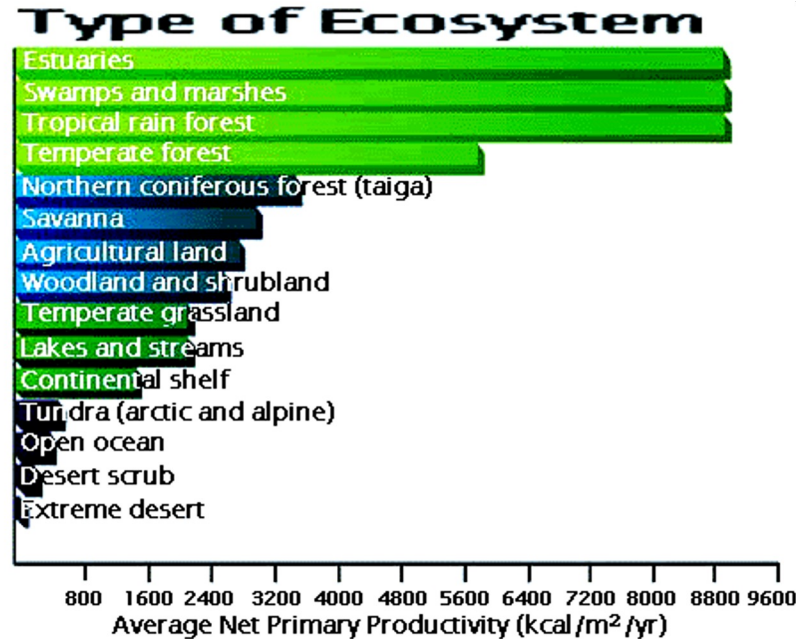
**D. M. Karl, R. Letelier, D. Hebel, L. Tupas,  
J. Dore, J. Christian & C. Winn**

SOEST, Department of Oceanography, University of Hawaii, Honolulu,  
Hawaii 96822, USA

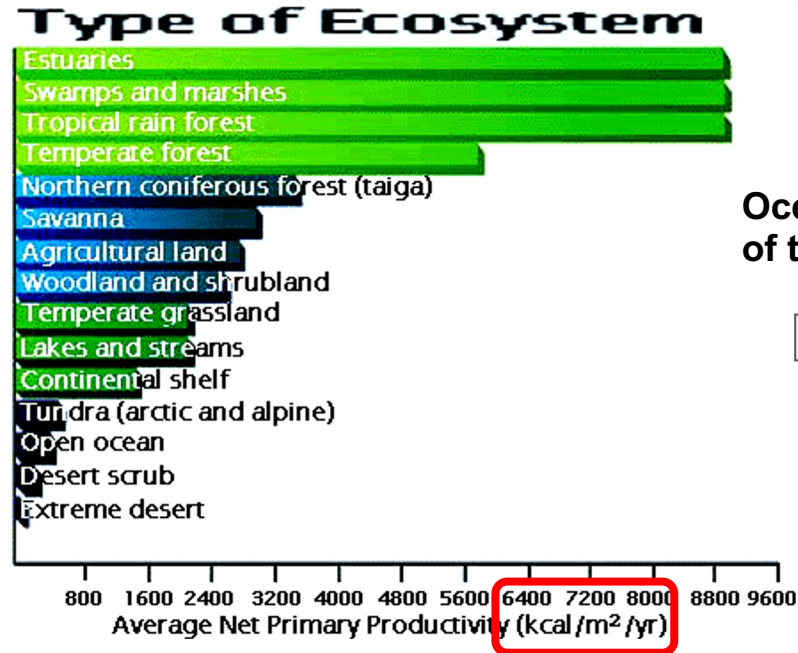
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SUBTROPICAL ocean gyres are considered to be the marine analogues of terrestrial deserts because of chronic nutrient depletion and low standing stocks of organisms<sup>1</sup>. Despite their presumed low rates of primary and export production, oligotrophic habitats contribute significantly to global productivity because of their large extent<sup>2</sup>. Therefore, even small changes in ecosystem production can produce large effects in the global carbon cycle. The North Pacific subtropical gyre has generally been thought to support a homogeneous, stable biological community<sup>3,4</sup>, but recent investigations have suggested instead that the ecosystem of this gyre is temporally and spatially variable<sup>5–7</sup>. The causes of this variability are not well understood. Here we present evidence of a major change in the structure and productivity of the pelagic ecosystem in the subtropical North Pacific Ocean, an effect that we attribute to the 1991–92 El Niño–Southern Oscillation (ENSO) event. Decreased upper-ocean mixing and a change in circulation resulted in an increased abundance and activity of nitrogen-fixing microorganisms and a shift from a primarily nitrogen-limited to a primarily phosphorus-limited habitat with attendant changes in total and export production and in nutrient cycling pathways and rates.

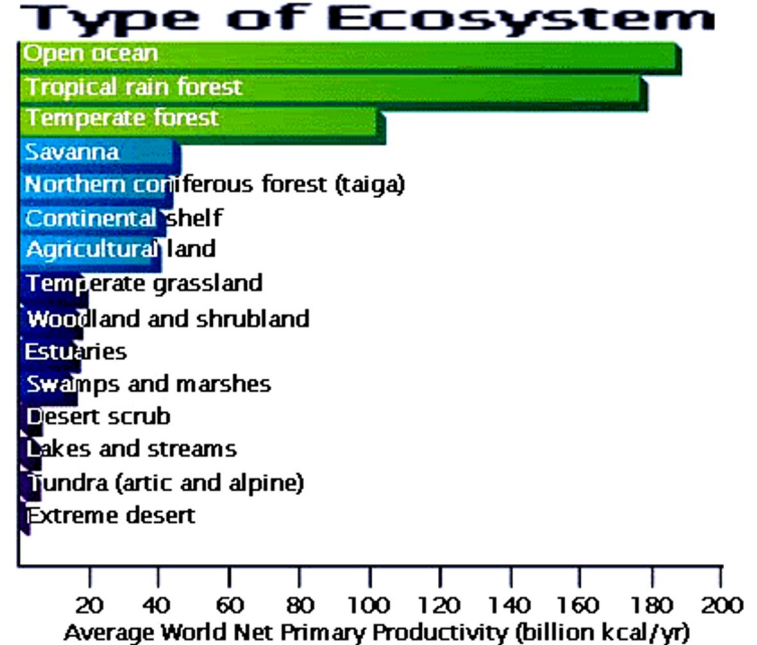
Answer: Open-ocean areas are oligotrophic (small primary production rates)



Answer: However, open-ocean are extremely large areas



Ocean occupies 71%  
of the Earth surface!



**Open Ocean ~ 26%, Rain Forest ~ 24% - 50:50 contribution**

# North Pacific Gyre Study Case: the complexity of the interconnected Earth system

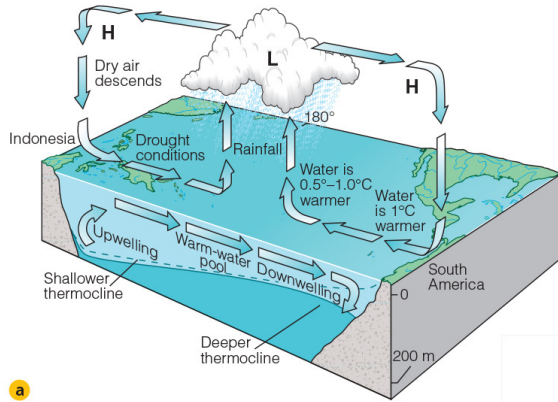


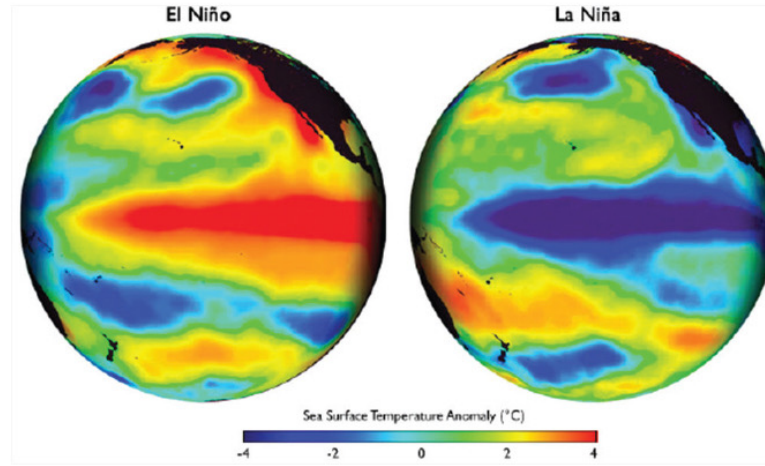
Figure 9.24 An El Niño year.

- a When the Southern Oscillation develops, the trade winds diminish and then reverse, leading to an eastward movement of warm water along the equator. The surface waters of the central and eastern Pacific become warmer, and storms over land may increase.



<https://encounteredu.com/multimedia/images/where-is-the-north-pacific-gyre>

ENSO has global impact



Steve Albers, NOAA

The authors analyzed several environmental parameters near Hawaii (middle of the North Pacific Gyre):

| TABLE 1 Oceanic habitat and ecosystem changes measured at station ALOHA (22° 45' N, 158° W) |                        |                        |               |
|---------------------------------------------------------------------------------------------|------------------------|------------------------|---------------|
| Parameter*                                                                                  | 1989–90 period†        | 1991–92 period         | Significance‡ |
| Total incident solar irradiance§ (MJ m <sup>-2</sup> )                                      | 1.08 × 10 <sup>4</sup> | 1.02 × 10 <sup>4</sup> | NS            |
| Mean sea surface temperature   (°C)                                                         | 25.14 ± 1.36 (667)     | 25.16 ± 1.56 (655)     | NS            |
| Surface-ocean dissolved nutrients (mg m <sup>-2</sup> )                                     |                        |                        |               |
| Total dissolved nitrogen*                                                                   | 3,529 ± 456 (15)       | 4,000 ± 315 (21)       | P < 0.001     |
| Total dissolved phosphorus**                                                                | 437 ± 43 (15)          | 481 ± 58 (21)          | P < 0.01      |
| Surface-ocean suspended particulate matter (mg m <sup>-2</sup> )                            |                        |                        |               |
| Chlorophyll a                                                                               | 4.5 ± 1.3 (17)         | 3.9 ± 1.1 (21)         | NS            |
| Carbon                                                                                      | 1.290 ± 278 (14)       | 1.397 ± 266 (21)       | NS            |
| ATP                                                                                         | 1.63 ± 0.31 (16)       | 1.30 ± 0.26 (19)       | P < 0.01      |
| Surface-ocean suspended particulate matter ratios**                                         |                        |                        |               |
| PC:PN (molar)                                                                               |                        |                        |               |
| Mean                                                                                        | 8.3 ± 3.8 (13)         | 7.3 ± 1.4 (20)         | NS            |
| Median                                                                                      | 7.1 (13)               | 6.9 (20)               |               |
| 'Redfield'                                                                                  | 6.6                    | 6.6                    |               |
| PC:PP (molar)                                                                               |                        |                        |               |
| Mean                                                                                        | 112 ± 32 (14)          | 136 ± 24 (21)          | P < 0.02      |
| Median                                                                                      | 111 (14)               | 135 (21)               |               |
| 'Redfield'                                                                                  | 106                    | 106                    |               |
| PN:PP (molar)                                                                               |                        |                        |               |
| Mean                                                                                        | 14.2 ± 3.3 (13)        | 19.3 ± 5.2 (20)        | P < 0.01      |
| Median                                                                                      | 13.6 (13)              | 18.7 (20)              |               |
| 'Redfield'                                                                                  | 16                     | 16                     |               |
| PC:P-ATP ratio (by weight)                                                                  | 822 ± 251 (14)         | 1,116 ± 300 (19)       | P < 0.01      |
| Downward flux of particulate matter†† (mg m <sup>-2</sup> d <sup>-1</sup> )                 |                        |                        |               |
| Carbon                                                                                      | 35.5 ± 11.5 (18)       | 24.4 ± 9.5 (18)        | P < 0.01      |
| Nitrogen                                                                                    | 4.99 ± 1.62 (18)       | 3.75 ± 1.53 (18)       | P < 0.02      |
| Phosphorus                                                                                  | 0.49 ± 0.19 (18)       | 0.36 ± 0.15 (18)       | P < 0.02      |
| Trichodesmium abundance estimates§§                                                         |                        |                        |               |
| % of total particulate carbon                                                               | 5.8 ± 2.7 (9)          | 11.3 ± 3.3 (13)        | P < 0.001     |
| % of total chl a                                                                            | 8.1 ± 6.4 (9)          | 17.1 ± 7.0 (13)        | P < 0.005     |

**Group activity # 1: Analyze the data provided in Table 1, Karl et al. 1995.**

1. Which trends can you identify?
2. Which parameters directly impact primary production?



ENSO period

TABLE 1 Oceanic habitat and ecosystem changes measured at station ALOHA (22° 45' N, 158° W)

| Parameter*                                                                  | 1989–90 period†        | 1991–92 period         | Significance‡ |
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| <i>Trichodesmium</i> abundance estimates§§                                  |                        |                        |               |
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| % of total chl a                                                            | 8.1 ± 6.4 (9)          | 17.1 ± 7.0 (13)        | P < 0.005     |

ENSO period



## 1. Which trends can you identify?

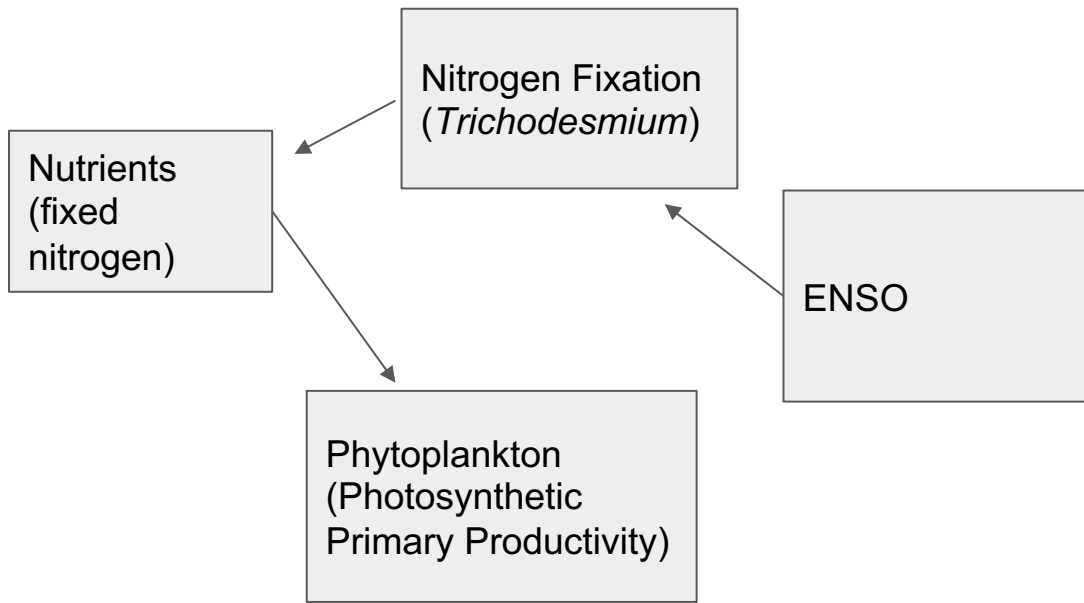
**Answer:** Light, temperature, particulate matter do not change; nutrient availability and nitrogen fixation organisms increase; and particulate nutrient ratios change during El Nino. Downward flux of particulate matter decreases.

## 2. Which parameters directly impact primary production?

**Answer:** In this case only nutrient availability. However, organisms that can change the nutrient abundance also play a role.

## Group activity #2

1. Based on your analysis of what parameters change during an El Nino event, build a box and arrow model and predict primary production change in the North Pacific gyre during the El Nino event, provide reasoning for your prediction.
2. Use the model to explain what do you expect to happen shortly after El Nino terminates or if a La Nina develops instead?



Primary production increases due to higher nutrient nitrogen availability fixed by the *Trichodesmium*

